

Software Process Improvement –SEPI-471



PDCA cycle

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Agenda

- PDCA Cycle

PDCA Cycle



PDCA Cycle

PDCA Cycle is a four step model is used for following

Continuous Improvement: PDCA is a cycle that repeats, allowing for continuous improvement over time.

Problem Solving: PDCA is an effective method for systematically solving problems and addressing challenges.

Quality Management: PDCA is widely used in quality management systems to enhance processes and outcomes.

Iterative Approach: PDCA embraces an iterative approach, learning from each cycle to inform the next.

PDCA Cycle

Plan:

It consists of following actions or activities

1. Problem Identification:

Clearly define the problem or opportunity for improvement. It involves analyzing data, feedback, or identifying a gap in performance

2. Objective Setting

Establish specific, measurable, achievable, relevant, and time-bound (SMART) objectives. Objectives provide a clear target for improvement.

3. Data Collection

Gather relevant data and information to understand the current state of the process or system.

4. Scope Definition

Clearly define the scope and boundaries of the improvement project

5. Action plan

Develop a detailed plan outlining the actions needed to achieve the objectives. This may include changes to processes, training, resource allocation, etc.

PDCA Cycle

Do:

It consists of following actions or activities

1. **Implementation:** Execute the plan as designed during the planning phase. This could involve making changes to processes, introducing new technologies, or implementing new procedures.
2. **Communication:** Ensure that all relevant stakeholders are informed and involved in the changes.
3. **Documentation:** Record any changes made during the implementation phase.
4. **Training:** If the changes involve new processes or technologies, provide necessary training to personnel.

PDCA Cycle

Check:

It consists of following actions or activities

Data Analysis: Analyze the data collected during the implementation phase to identify any variances or deviations.

Evaluation: Assess the effectiveness of the implemented changes in addressing the identified problem or achieving the set objectives.

Feedback: Collect feedback from stakeholders and team members involved in the process.

PDCA Cycle

Act:

It consists of following actions or activities

1. **Standardization:** If the results are satisfactory, standardize the new process or practices.
2. **Revisions:** If improvements are needed, revise the plan accordingly.
3. **Implementation:** Implement the revised plan. This begins a new iteration of the PDCA cycle, starting again with the Plan phase.
4. **Documentation and Learning:** Document the actions taken and the lessons learned during the PDCA cycle.

Example: Improving Customer Service in a Call Center:

Plan:

Objective: Reduce customer wait times by 20% within the next quarter.

Activities:

- Identify customer service data, including average wait times and customer satisfaction scores.
- Define the scope to focus on specific call center processes.
- Set the SMART objective of reducing wait times by 20%.
- Develop a plan that includes process improvements, additional training, and updated technology.

Do:

Objective: Implement the plan to improve customer service processes.

Activities:

- Roll out new training programs for call center representatives.
- Implement updated call routing technology.
- Monitor and adjust staffing levels to accommodate peak call times.

check:

Objective: Assess the impact of the implemented changes on wait times and customer satisfaction.

Activities:

- Collect data on wait times and customer satisfaction scores.
- Compare actual results with the targeted 20% reduction.
- Analyze customer feedback and identify areas for further improvement.

PDCA Cycle

Act:

Objective: Take actions based on the assessment to further improve customer service.

Activities:

- If the 20% reduction is achieved, standardize the new processes.
- If not, revise the plan based on the analysis and customer feedback.
- Implement the revised plan.
- Continue the PDCA cycle for continuous improvement.

PDCA Cycle-Example

You are the quality manager for a development company that produces widgets. Recently, you have noticed an increase in customer complaints about the quality of the widgets. You suspect that the problem may be related to the assembly process, but you are not sure. You decide to use the Plan-Do-Check-Act (PDCA) cycle to address the issue. How you will apply PDCA cycle to satisfy your customers.